

AEROFIT Business case Study

~ by Shivaling

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- **About Aerofit**

Aerofit is a leading brand in the field of fitness equipment. Aerofit provides a product range including machines such as treadmills, exercise bikes, gym equipment, and fitness accessories to cater to the needs of all categories of people.

- **Business Problem**

The market research team at AeroFit wants to identify the characteristics of the target audience for each type of treadmill offered by the company, to provide a better recommendation of the treadmills to the new customers. The team decides to investigate whether there are differences across the product with respect to customer characteristics.

1. Perform descriptive analytics to create a customer profile for each AeroFit treadmill product by developing appropriate tables and charts.
2. For each AeroFit treadmill product, construct two-way contingency tables and compute all conditional and marginal probabilities along with their insights/impact on the business.

- **Dataset**

The company collected the data on individuals who purchased a treadmill from the AeroFit stores during the prior three months. The dataset has the following features:

Dataset link: [Aerofit treadmill.csv](#)

Product	KP281, KP481, or KP781
Purchased:	
Age:	In years
Gender:	Male/Female
Education:	In years
MaritalStatus:	Single or partnered
Usage:	The average number of times the customer plans to use the treadmill each week.
Income:	Annual income (in \$)
Fitness:	Self-rated fitness on a 1-to-5 scale, where 1 is the poor shape and 5 is the excellent shape.
Miles:	The average number of miles the customer expects to walk/run each week

- **Product Portfolio:**

- The KP281 is an entry-level treadmill that sells for \$1,500.
- The KP481 is for mid-level runners that sell for \$1,750.
- The KP781 treadmill is having advanced features that sell for \$2,500.

Google Colab Link:

<https://colab.research.google.com/drive/1OcIrl-L500EnNdsMbcPqS2wHuWbb9yt?usp=sharing>

▼ *AEROFIT Business Case Study*

▼ Phase 1: Importing aerofit_treadmill.csv Dataset

```
1 !gdown 1EmyoBhy3oU2RHMBsog4V06qs5yPBqmH5 -O aerofit.csv

Downloading...
From: https://drive.google.com/uc?id=1EmyoBhy3oU2RHMBsog4V06qs5yPBqmH5
To: /content/aerofit.csv
100% 7.28k/7.28k [00:00<00:00, 17.3MB/s]
```

▼ Phase 2: Importing Required Libraries

```
1 import numpy as np
2 import pandas as pd
3 import seaborn as sns
4 import matplotlib.pyplot as plt
```

▼ Phase 3: Reading the Dataset into Data Frame

```
1 atm = pd.read_csv('aerofit.csv')
2 atm
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47
...	...	...	...	...	...	...	...	...	...
175	KP781	40	Male	21	Single	6	5	83416	200
176	KP781	42	Male	18	Single	5	4	89641	200
177	KP781	45	Male	16	Single	5	5	90886	160
178	KP781	47	Male	18	Partnered	4	5	104581	120
179	KP781	48	Male	18	Partnered	4	5	95508	180

180 rows x 9 columns

```
1 atm.head()
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

```
1 atm.tail()
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
175	KP781	40	Male	21	Single	6	5	83416	200
176	KP781	42	Male	18	Single	5	4	89641	200
177	KP781	45	Male	16	Single	5	5	90886	160
178	KP781	47	Male	18	Partnered	4	5	104581	120
179	KP781	48	Male	18	Partnered	4	5	95508	180

▼ Phase 4: Understanding the Dataset

```
1 atm.shape
```

(180, 9)

1 # The Aerofit Dataset consists of 180 Rows and 9 Columns

1 # Columns in the Dataset
2 atm.columns

```
Index(['Product', 'Age', 'Gender', 'Education', 'MaritalStatus', 'Usage',  
      'Fitness', 'Income', 'Miles'],  
      dtype='object')
```

1 atm.info()

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 180 entries, 0 to 179  
Data columns (total 9 columns):  
#   Column                Non-Null Count  Dtype  
---  ---  
0   Product                180 non-null   object  
1   Age                    180 non-null   int64  
2   Gender                 180 non-null   object  
3   Education              180 non-null   int64  
4   MaritalStatus          180 non-null   object  
5   Usage                  180 non-null   int64  
6   Fitness                180 non-null   int64  
7   Income                 180 non-null   int64  
8   Miles                  180 non-null   int64  
dtypes: int64(6), object(3)  
memory usage: 12.8+ KB
```

1 # The dataset has 3 columns which are Objects, i.e. Product, Gender, MaritalStatus, they can be converted to categories for further D

```
1 from pandas.core.indexes import category  
2 cat_cols = ['Product', 'Gender', 'MaritalStatus']  
3 for col in cat_cols:  
4     atm[col] = atm[col].astype('category')
```

1 atm.describe(include = 'all')

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
count	180	180.000000	180	180.000000	180	180.000000	180.000000	180.000000	180.000000
unique	3	NaN	2	NaN	2	NaN	NaN	NaN	NaN
top	KP281	NaN	Male	NaN	Partnered	NaN	NaN	NaN	NaN
freq	80	NaN	104	NaN	107	NaN	NaN	NaN	NaN
mean	NaN	28.788889	NaN	15.572222	NaN	3.455556	3.311111	53719.577778	103.194444
std	NaN	6.943498	NaN	1.617055	NaN	1.084797	0.958869	16506.684226	51.863605
min	NaN	18.000000	NaN	12.000000	NaN	2.000000	1.000000	29562.000000	21.000000
25%	NaN	24.000000	NaN	14.000000	NaN	3.000000	3.000000	44058.750000	66.000000
50%	NaN	26.000000	NaN	16.000000	NaN	3.000000	3.000000	50596.500000	94.000000
75%	NaN	33.000000	NaN	16.000000	NaN	4.000000	4.000000	58668.000000	114.750000
max	NaN	50.000000	NaN	21.000000	NaN	7.000000	5.000000	104581.000000	360.000000

Phase 5: Handle Missing Values

1 atm.isnull().sum()

	0
Product	0
Age	0
Gender	0
Education	0
MaritalStatus	0
Usage	0
Fitness	0
Income	0
Miles	0

dtype: int64

```
1 # Tha Dataset is clean, Any Column or Row does not contains any null or missing values.
```

```
1 atm.duplicated().sum()
```

```
np.int64(0)
```

```
1 # Tha Dataset does not contains any repeating records.
```

Phase 6: Non Graphical Analysis

```
1 atm['Age'].min(), atm['Age'].max()
```

```
(18, 50)
```

```
1 # The youngest Customer is of the age 18 and the oldest is aged 50.
```

```
1 age_bins = [13, 19, 29, 39, 49, 59]
2 age_labels = ['Teenage', '20s', '30s', '40s', '50s']
3 atm['Age_Group'] = pd.cut(atm['Age'], bins = age_bins, labels = age_labels)
4 atm.head()
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles	Age_Group
0	KP281	18	Male	14	Single	3	4	29562	112	Teenage
1	KP281	19	Male	15	Single	2	3	31836	75	Teenage
2	KP281	19	Female	14	Partnered	4	3	30699	66	Teenage
3	KP281	19	Male	12	Single	3	3	32973	85	Teenage
4	KP281	20	Male	13	Partnered	4	2	35247	47	20s

```
1 atm['Age_Group'].value_counts()
```

count	
Age_Group	
20s	108
30s	50
40s	16
Teenage	5
50s	1

```
dtype: int64
```

```
1 # Aerofit has attracted customers aged in their 20's dominantly, it states that the
2 # most health consious customers who use Aerofit Products are under 30 in age.
```

```
1 atm['Product'].value_counts()
```

count	
Product	
KP281	80
KP481	60
KP781	40

```
dtype: int64
```

```
1 # Majority of customers have bought KP281 Product, which suggests KP281 is mostly preferred by customers, as It is entry level and is
2 # whereas KP781 is least bought by customers, irrespective of the Gender, Marital Status, It does provides advanced features, but may
3 # as it id for the KP281.
```

```
1 atm['Gender'].value_counts()
```

count	
Gender	
Male	104
Female	76

```
dtype: int64
```

1 # Aerofit products attracted Male Gender dominant Customers than Female ones, across all types of Treadmill products.

```
1 atm['MaritalStatus'].value_counts()
```

count	
MaritalStatus	
Partnered	107
Single	73

dtype: int64

1 # Arofit products attract married, co living customers more than single ones.

```
1 atm['Product'].value_counts(normalize = True)
```

proportion	
Product	
KP281	0.444444
KP481	0.333333
KP781	0.222222

dtype: float64

1 # Almost 45% of Arofit Customers preferred KP281, 33% preferred KP481 and 22% preferred KP781

```
1 atm['Gender'].value_counts(normalize = True)
```

proportion	
Gender	
Male	0.577778
Female	0.422222

dtype: float64

1 # Male Customers (58%) use Aerofit Products more than Female ones (42%), across all age, marital status, fitness category.
2 # Aerofit has attracted both Male and female customers approximately. But Male dominate in usage.

```
1 atm['MaritalStatus'].value_counts(normalize = True)
```

proportion	
MaritalStatus	
Partnered	0.594444
Single	0.405556

dtype: float64

1 # 59% of Aerofit Customers are partnered, and remaing 41% are Single.

1 # For all Numerical Columns, Let us Describe them to get a idea of the Statistics

```
1 atm.head(1)
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles	Age_Group
0	KP281	18	Male	14	Single	3	4	29562	112	Teenage

```
1 atm[['Age', 'Education', 'Usage', 'Fitness', 'Income', 'Miles']].describe()
```

	Age	Education	Usage	Fitness	Income	Miles
count	180.000000	180.000000	180.000000	180.000000	180.000000	180.000000
mean	28.788889	15.572222	3.455556	3.311111	53719.577778	103.194444
std	6.943498	1.617055	1.084797	0.958869	16506.684226	51.863605
min	18.000000	12.000000	2.000000	1.000000	29562.000000	21.000000
25%	24.000000	14.000000	3.000000	3.000000	44058.750000	66.000000
50%	26.000000	16.000000	3.000000	3.000000	50596.500000	94.000000
75%	33.000000	16.000000	4.000000	4.000000	58668.000000	114.750000
max	50.000000	21.000000	7.000000	5.000000	104581.000000	360.000000

```

1 # From the above analysis, We can deduce the following:
2 # The customers Ages ranges from 18 to 50, which means all the targeted customers are Adults.
3 # The Mean 28.79 and median 26 for Age suggests that, there is not much of a variance
4 # of data in Age of customers, showing uniformity, less skewness in data arrangement.
5 # The average no. ofg times customers planned to use the Treadmills ranges from 2 to 7 times.
6 # Most of the customers plan to use more than 3 times in a week
7 # Fitness of customers in a scale of 1 - 5 suggests that, 75% of them have rated below 4,
8 # and correspondingly there no. of usage shall be high
9

```

```
1 atm['Fitness'].value_counts()
```

	count
Fitness	
3	97
5	31
2	26
4	24
1	2

```
dtype: int64
```

```

1 # Among Aerofit customers who have self Rated in scales of 1 to 5, Maximum no. of customers are rated 3, 97 customers, followed by ra
2 # There are 2 customers who are in Poor shape (rating 1) and 31 customers who are in excellent shape (rating 5).

```

```
1 pd.crosstab(index = [atm['Gender'], atm['Product']], columns = atm['Age'], margins = True)
```

		Age	18	19	20	21	22	23	24	25	26	27	...	41	42	43	44	45	46	47	48	50	All
Female	KP281		0	1	1	2	3	3	3	4	3	2	...	0	0	0	1	0	1	0	0	1	40
	KP481		0	0	1	1	0	3	2	5	2	0	...	0	0	0	0	0	0	0	0	0	29
	KP781		0	0	0	0	0	1	1	1	1	0	...	0	0	0	0	0	0	0	0	0	7
Male	KP281		1	2	1	2	1	5	2	3	4	1	...	1	0	1	0	0	0	1	0	0	40
	KP481		0	1	2	2	0	4	1	6	1	1	...	0	0	0	0	1	0	0	1	0	31
	KP781		0	0	0	0	3	2	3	6	1	3	...	0	1	0	0	1	0	1	1	0	33
All			1	4	5	7	7	18	12	25	12	7	...	1	1	1	1	2	1	2	2	1	180

```
7 rows x 33 columns
```

```

1 # The above crosstab shows the no. of customers acroos different age who use different aerofit Treadmill Products.
2 #

```

```
1 pd.crosstab(index = [atm['Product'], atm['Fitness']], columns = [atm['Gender'], atm['MaritalStatus']], margins = True)
```

	Gender	Female		Male		All
	MaritalStatus	Partnered	Single	Partnered	Single	
Product	Fitness					
KP281	1	0	0	1	0	1
	2	7	3	4	0	14
	3	18	8	13	15	54
	4	1	2	3	3	9
	5	1	0	0	1	2
KP481	1	0	1	0	0	1
	2	2	4	5	1	12
	3	12	6	13	8	39
	4	1	3	3	1	8
KP781	3	0	1	1	2	4
	4	0	1	5	1	7
	5	4	1	13	11	29
All		46	30	61	43	180

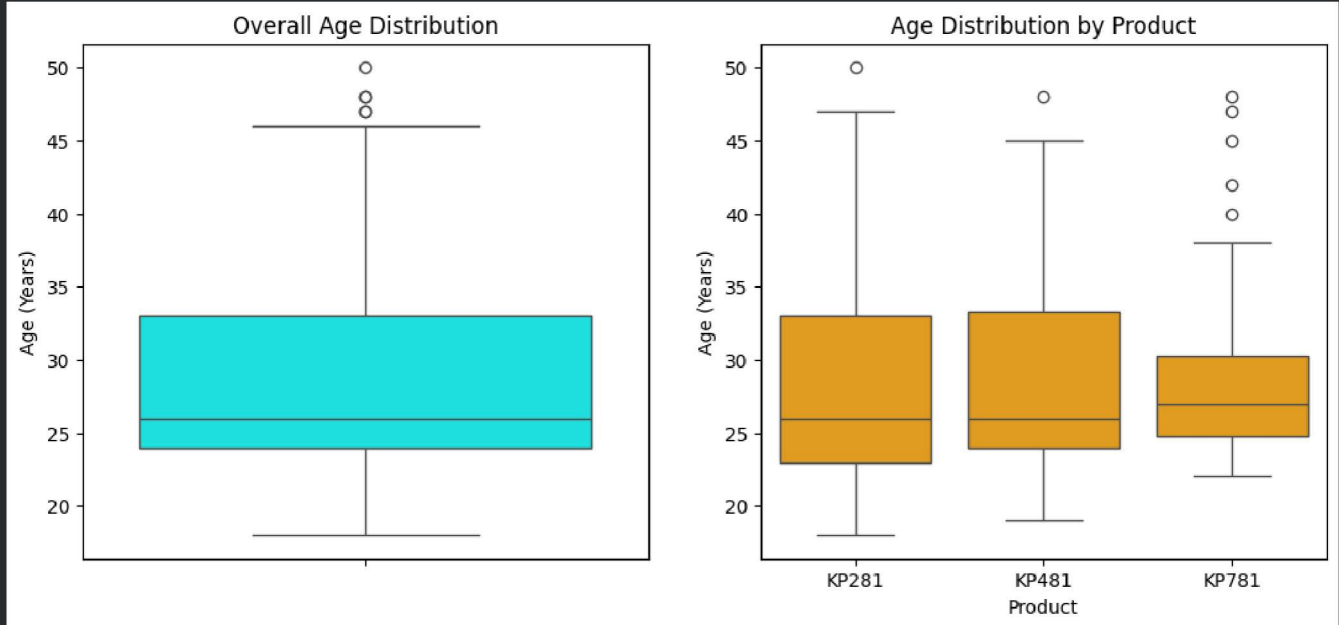
```
1 pd.crosstab(index = [atm['Gender'], atm['MaritalStatus']], columns = [atm['Product'], atm['Fitness']], margins = True)
```

	Product	KP281					KP481					KP781					All
	Fitness	1	2	3	4	5	1	2	3	4	3	4	5				
Gender	MaritalStatus																
Female	Partnered	0	7	18	1	1	0	2	12	1	0	0	4	46			
	Single	0	3	8	2	0	1	4	6	3	1	1	1	30			
Male	Partnered	1	4	13	3	0	0	5	13	3	1	5	13	61			
	Single	0	0	15	3	1	0	1	8	1	2	1	11	43			
All		1	14	54	9	2	1	12	39	8	4	7	29	180			

```
1 pd.crosstab(index = [atm['Gender'], atm['MaritalStatus']], columns = [atm['Fitness']], margins = True)
```

		Fitness	1	2	3	4	5	All
Gender	MaritalStatus							
Female	Partnered	0	9	30	2	5	46	
	Single	1	7	15	6	1	30	
Male	Partnered	1	9	27	11	13	61	
	Single	0	1	25	5	12	43	
All		2	26	97	24	31	180	

```
1 plt.figure(figsize = (12, 5))
2
3 plt.subplot(1,2,1)
4 abp = sns.boxplot(y = atm['Age'], color = 'cyan')
5
6 plt.title('Overall Age Distribution')
7 plt.ylabel('Age (Years)')
8
9 plt.subplot(1,2,2)
10 abbp = sns.boxplot(y = atm['Age'], x = atm['Product'], color = 'orange')
11 plt.title('Age Distribution by Product')
12 plt.xlabel('Product')
13 plt.ylabel('Age (Years)')
14
15 plt.show()
```



1 # From the above Boxplots, the overall Average for Customers age is around 26.

```
1 p_g = pd.crosstab(atm['Product'], atm['Gender'])
2 p_g
```

Gender	Female	Male
Product		
KP281	40	40
KP481	29	31
KP781	7	33

1 # The above analysis shows Genderwise Product purchase, KP281 is most common among both Genders.
 2 # We can say that KP281 is beginner friendly, entry level treadmill, lowest in Price i.e. \$1500 is mostly preferred by anyone.
 3 # Whereas, KP781 available with advanced features and is also costlier in price, is least preferred among any gender.

Phase 7: Graphical Analysis (Univariate)

```
1 atm.head(1)
```

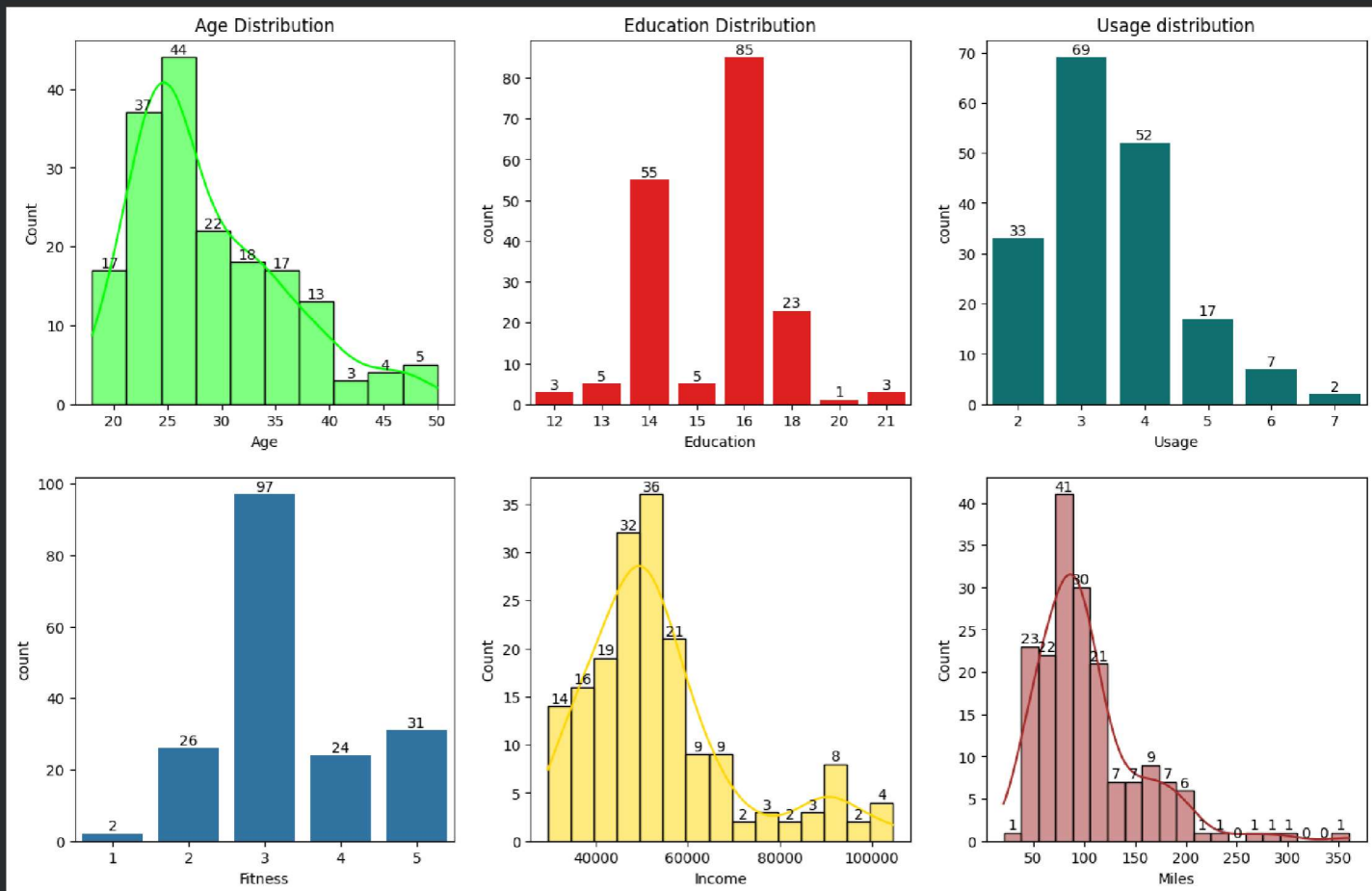
	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles	Age_Group
0	KP281	18	Male	14	Single	3	4	29562	112	Teenage

```
1 plt.figure(figsize = (16,10))
2
3 # Age Distribution of Customers
4 plt.subplot(2,3,1)
5 ax = sns.histplot(atm['Age'], bins = 10, color = 'lime', kde = True)
6 for container in ax.containers:
7     ax.bar_label(container)
8 plt.title('Age Distribution')
9
10 # Education Level Distribution of Customers
11 plt.subplot(2,3,2)
12 ax = sns.countplot(data = atm, x = 'Education', color = 'r')
13 for container in ax.containers:
14     ax.bar_label(container)
15 plt.title('Education Distribution')
16
17 # Usage per week Distribution of customers
18 plt.subplot(2,3,3)
19 ax = sns.countplot(x = atm['Usage'], color = 'teal')
20 for container in ax.containers:
21     ax.bar_label(container)
22 plt.title('Usage distribution')
23
24 # Fitness Rating Distribution across all customers
25 plt.subplot(2,3, 4)
26 ax = sns.countplot(data = atm, x='Fitness')
27 for container in ax.containers:
28     ax.bar_label(container)
29
30 # Income Distribution of customers
31 plt.subplot(2, 3,5)
```

```

32 ax = sns.histplot(x = atm['Income'], color = 'gold', kde = True, bins = 15)
33 for container in ax.containers:
34     ax.bar_label(container)
35
36 # Average number of Miles distribution across customers
37 plt.subplot(2,3,6)
38 ax = sns.histplot(x = atm['Miles'], color = 'brown', kde = True)
39 for container in ax.containers:
40     ax.bar_label(container)
41
42 plt.show()

```



```

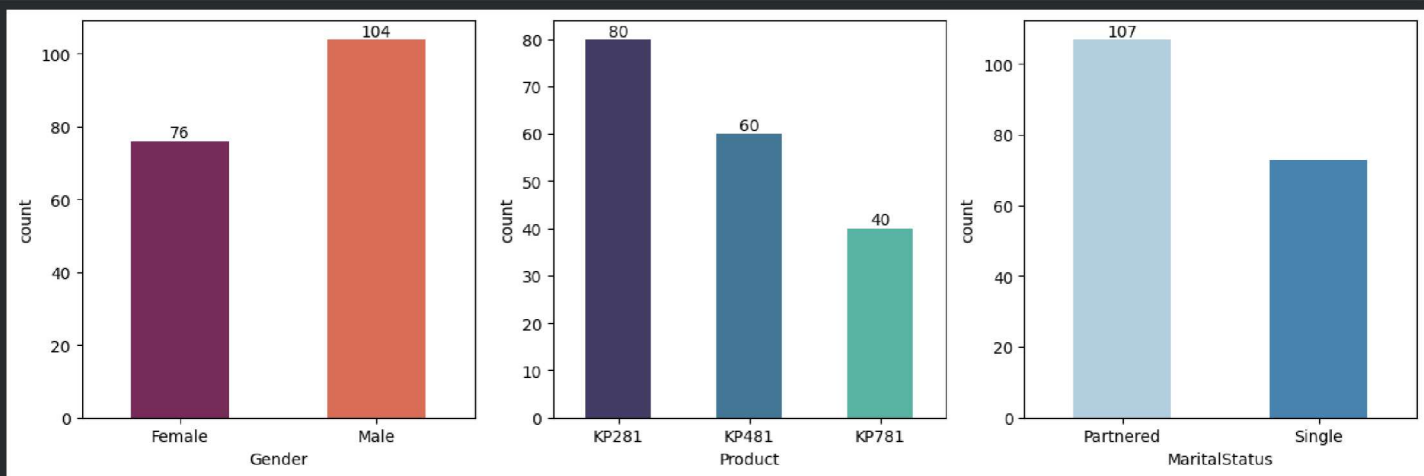
1 # The above subplots show the distribution analysis for numerical columns
2 # Age, Usage, Income and Miles distributions are Right Skewed,
3 # whereas Fitness Distribution is Left Skewed.

```

```

1 plt.figure(figsize = (15, 4.5))
2
3 plt.subplot(1,3,1)
4 ax = sns.countplot(x = atm['Gender'], hue = atm['Gender'], palette = 'rocket', width = 0.5)
5 for container in ax.containers:
6     ax.bar_label(container)
7
8 plt.subplot(1,3,2)
9 ax = sns.countplot(x = atm['Product'], hue = atm['Product'], palette = 'mako', width = 0.5)
10 for container in ax.containers:
11     ax.bar_label(container)
12
13 plt.subplot(1,3,3)
14 ax = sns.countplot(x = atm['MaritalStatus'], hue = atm['MaritalStatus'], palette = 'Blues', width = 0.5)
15 for container in ax.containers:
16     ax.bar_label(container)
17
18 plt.show()

```



1 # In Products, KP281 with lowest Price, beginner friendly, entry level dominates the treadmill product category
 2 # Among Marital Status, Partnered dominates among the aerofit Product Users
 3 # And, Male users dominate over female users across all category.

1 Start coding or [generate](#) with AI.

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Phase 8: Outliers

```
1 atm.head(1)
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles	Age_Group
0	KP281	18	Male	14	Single	3	4	29562	112	Teenage

```
1 # Let us summarize and check mean and median for numerical columns:
2
3 summary = atm[['Age', 'Education', 'Usage', 'Income', 'Fitness', 'Miles']].describe().T
4
5 summary[['mean', '50%']]
```

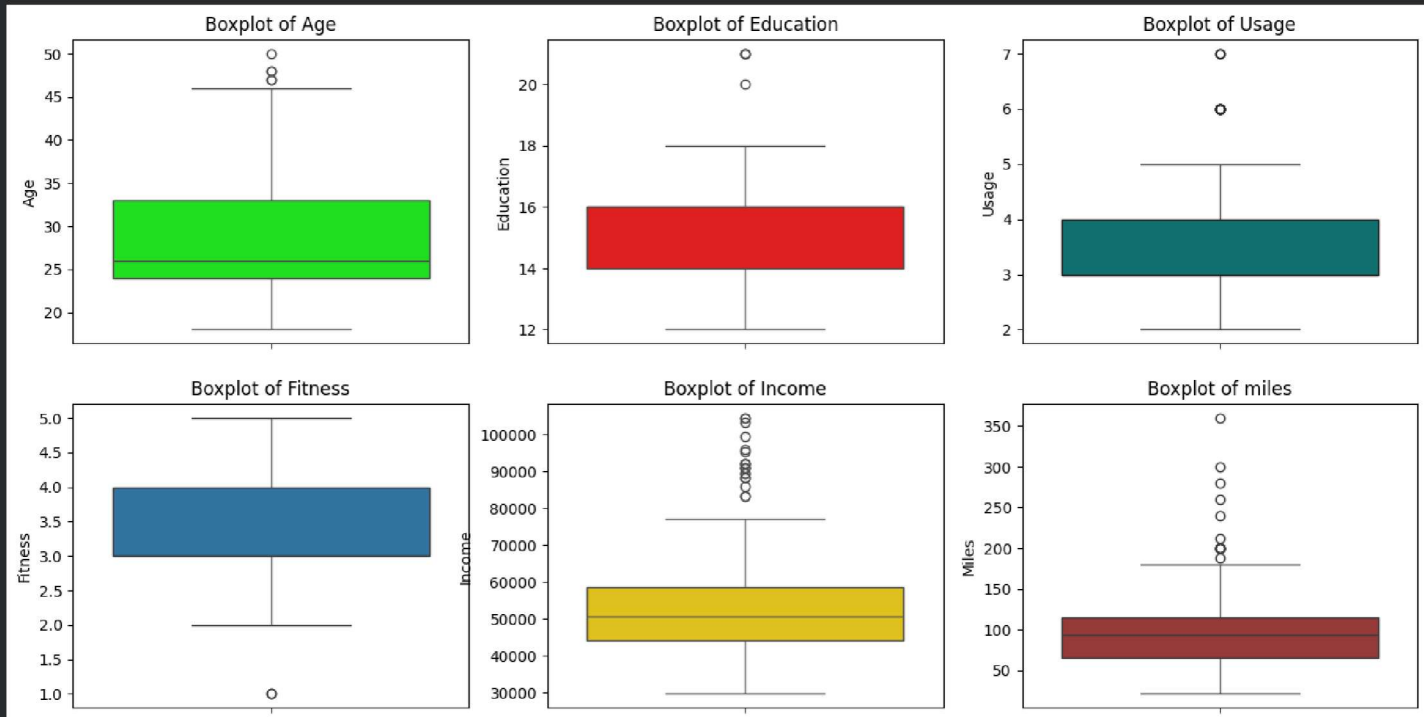
	mean	50%
Age	28.788889	26.0
Education	15.572222	16.0
Usage	3.455556	3.0
Income	53719.577778	50596.5
Fitness	3.311111	3.0
Miles	103.194444	94.0

```
1 plt.figure(figsize = (16, 8))
2
3 plt.subplot(2,3,1)
4 sns.boxplot(y = atm['Age'], color = 'lime')
5 plt.xticks(rotation = 45)
6 plt.title('Boxplot of Age')
7
8 plt.subplot(2,3,2)
9 sns.boxplot(y = atm['Education'], color = 'r')
10 plt.xticks(rotation = 45)
11 plt.title('Boxplot of Education')
12
13 plt.subplot(2,3,3)
14 sns.boxplot(y = atm['Usage'], color = 'teal')
15 plt.xticks(rotation = 45)
16 plt.title('Boxplot of Usage')
17
```

```

18 plt.subplot(2,3,4)
19 sns.boxplot(y = atm['Fitness'])
20 plt.xticks(rotation = 45)
21 plt.title('Boxplot of Fitness')
22
23 plt.subplot(2,3,5)
24 sns.boxplot(y = atm['Income'], color = 'gold')
25 plt.xticks(rotation = 45)
26 plt.title('Boxplot of Income')
27
28 plt.subplot(2,3,6)
29 sns.boxplot(y = atm['Miles'], color = 'brown')
30 plt.xticks(rotation = 45)
31 plt.title('Boxplot of miles')
32 plt.show()

```



```

1 # From the summary tabel and above Boxplots we can see Outliers are prominently more in Income and Miles.
2 # The difference between mean and median is greter in Income and Miles.
3 # Whereas in Age, Education, Usage and Fitness, there are less no. of Outliers
4 # which do not affect the skewness of Data relatively as it is with Income and Miles
5 # Let us detect the Outliers for Income, by further analysis:

```

```

1 i_s = atm['Income'].describe().T
2 i_s

```

	Income
count	180.000000
mean	53719.577778
std	16506.684226
min	29562.000000
25%	44058.750000
50%	50596.500000
75%	58668.000000
max	104581.000000

dtype: float64

```

1 atm.sort_values('Income', ascending=False).head()

```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles	Age_Group
178	KP781	47	Male	18	Partnered	4	5	104581	120	40s
174	KP781	38	Male	18	Partnered	5	5	104581	150	30s
168	KP781	30	Male	18	Partnered	5	4	103336	160	30s
169	KP781	30	Male	18	Partnered	5	5	99601	150	30s
171	KP781	33	Female	18	Partnered	4	5	95866	200	30s

1 # These High Income Values are far gretaeer than Mean and Median of Income, due to which there a right skewness in Data as seen in his

```
1 # Let Us identify the outliers using IQR
2
3 Q1 = atm['Income'].quantile(0.25)
4 Q3 = atm['Income'].quantile(0.75)
5 IQR = Q3 - Q1
6
7 lower = Q1 - 1.5 * IQR
8 upper = Q3 + 1.5 * IQR
9
10 print('IQR:', IQR)
11 print('Lower Bound:', lower)
12 print('Upper Bound:', upper)
```

IQR: 14609.25
Lower Bound: 22144.875
Upper Bound: 80581.875

```
1 # The Potential Outliners:
2 income_outliers = atm[(atm['Income'] < lower) | (atm['Income'] > upper)]
3
4 income_outliers
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles	Age_Group
159	KP781	27	Male	16	Partnered	4	5	83416	160	20s
160	KP781	27	Male	18	Single	4	3	88396	100	20s
161	KP781	27	Male	21	Partnered	4	4	90886	100	20s
162	KP781	28	Female	18	Partnered	6	5	92131	180	20s
164	KP781	28	Male	18	Single	6	5	88396	150	20s
166	KP781	29	Male	14	Partnered	7	5	85906	300	20s
167	KP781	30	Female	16	Partnered	6	5	90886	280	30s
168	KP781	30	Male	18	Partnered	5	4	103336	160	30s
169	KP781	30	Male	18	Partnered	5	5	99601	150	30s
170	KP781	31	Male	16	Partnered	6	5	89641	260	30s
171	KP781	33	Female	18	Partnered	4	5	95866	200	30s
172	KP781	34	Male	16	Single	5	5	92131	150	30s
173	KP781	35	Male	16	Partnered	4	5	92131	360	30s
174	KP781	38	Male	18	Partnered	5	5	104581	150	30s
175	KP781	40	Male	21	Single	6	5	83416	200	40s
176	KP781	42	Male	18	Single	5	4	89641	200	40s
177	KP781	45	Male	16	Single	5	5	90886	160	40s
178	KP781	47	Male	18	Partnered	4	5	104581	120	40s
179	KP781	48	Male	18	Partnered	4	5	95508	180	40s

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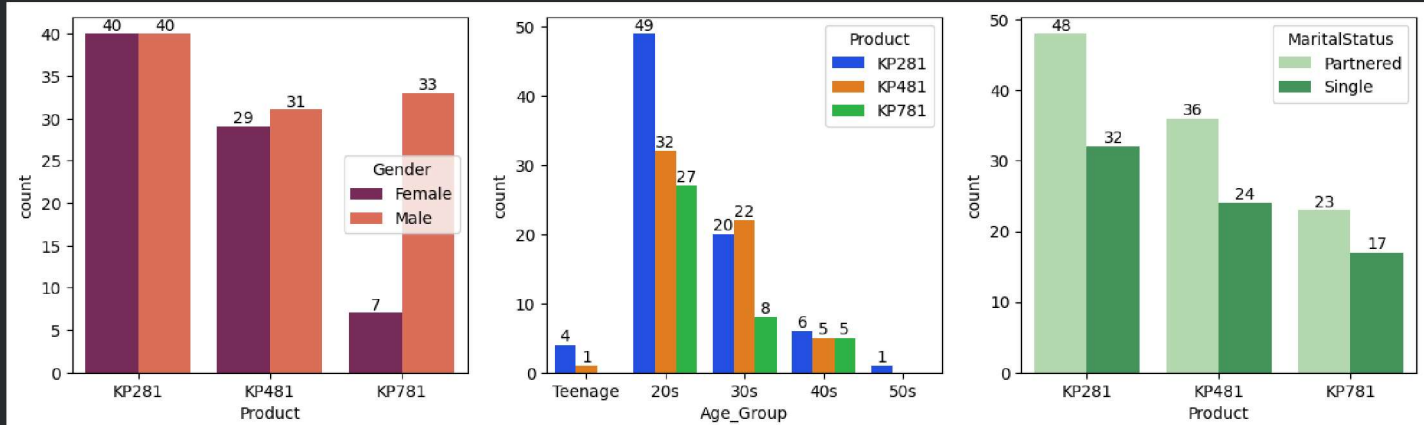
Phase 9: Bi-Variate Analysis

```
1 plt.figure(figsize = (15, 4))
2
3 plt.subplot(1,3,1)
4 ax = sns.countplot(data = atm, x = 'Product', hue = 'Gender', palette = 'rocket')
5 for container in ax.containers:
6     ax.bar_label(container)
7
8 plt.subplot(1,3,2)
9 ax = sns.countplot(data = atm, x = 'Age_Group', hue = 'Product', palette = 'bright')
10 for container in ax.containers:
```

```

11 ax.bar_label(container)
12
13 plt.subplot(1,3,3)
14 ax = sns.countplot(data = atm, x = 'Product', hue = 'MaritalStatus', palette = 'Greens')
15 for container in ax.containers:
16     ax.bar_label(container)
17
18

```



1 Start coding or [generate](#) with AI.

1 # In any category of Product, Male buyers/users are dominant than female.
2 # For KP781 which is the premium model, Male users are almost 4.5 times greater than female
3 # Which suggest advanced features attract Males irrespective of the price of model.

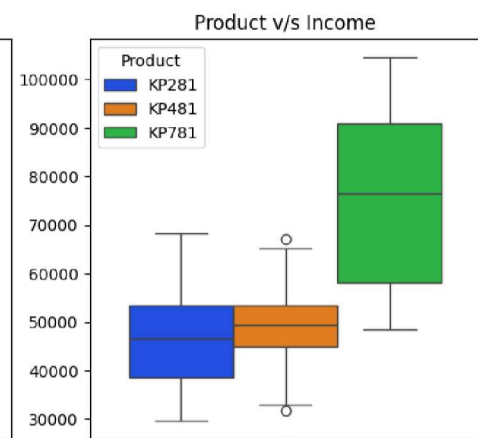
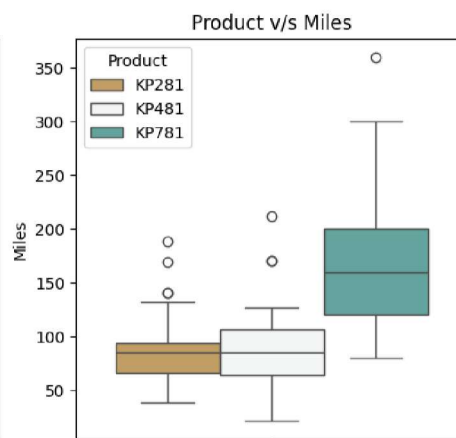
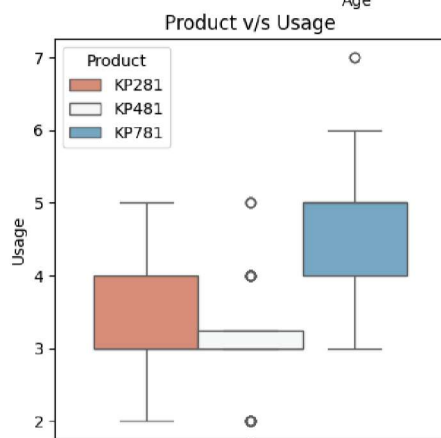
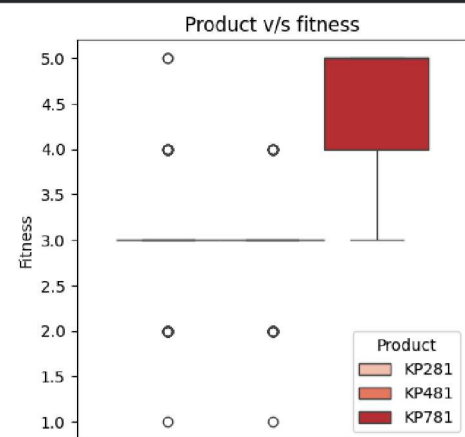
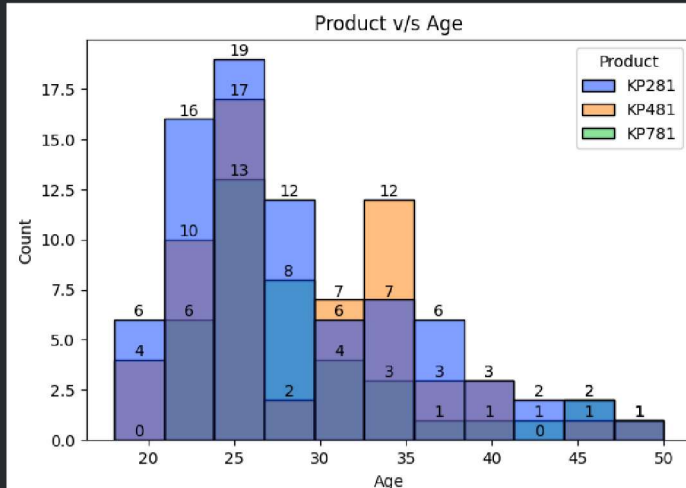
1 # Teenagers and Senior are least in customers for aerofit treadmills.
2 # The Young People who are in their 20s use aerofit treadmills and they prefer entry level model, cheapest i.e KP281.
3 # Even the premium Treadmill which is KP781, is most preferred by Young Users, because of its Advanced feature, which is mostly proba

1 # marital Status does affects the usage of Treadmills in any Product category.
2 # Partnered Users are dominant across all three categories of product

```

1 plt.figure(figsize = (15, 10))
2
3 plt.subplot(2,2,1)
4 ax = sns.histplot(data = atm, x = 'Age', hue = 'Product', palette = 'bright')
5 for container in ax.containers:
6     ax.bar_label(container)
7 plt.title('Product v/s Age')
8
9 plt.subplot(2,3,6)
10 ax = sns.boxplot(data = atm, y = 'Income', hue = 'Product', palette = 'bright')
11 plt.title('Product v/s Income')
12
13 plt.subplot(2,3,3)
14 ax = sns.boxplot(data = atm, y = 'Fitness', hue = 'Product', palette = 'Reds')
15 plt.title('Product v/s fitness')
16
17 plt.subplot(2,3,4)
18 ax = sns.boxplot(data = atm, y = 'Usage', hue = 'Product', palette = 'RdBu')
19 plt.title('Product v/s Usage')
20
21 plt.subplot(2,3,5)
22 ax = sns.boxplot(data = atm, y = 'Miles', hue = 'Product', palette = 'BrBG')
23 plt.title('Product v/s Miles')
24
25 plt.show()

```



- 1 # Younger Customers 20-30, dominate in buying Premium, advanced treadmill.
- 2 # Fitness Level is a enhancer in usage of Products, higher fitness rated customers tend to buy advanced models
- 3 # Customers who plan to use products atleast 5 times a week tend to buy advanced model and are ready to spend more money, for hassle
- 4 # Customers who plan to use products atmost 5 times a week tend to buy entry level model
- 5 # The customers expecting to wak/run less than 150 miles/week tend to stick with entry level model, whereas those who expect to walk/
- 6 # tend to use a dvanced featuristic model
- 7 # customers income directly affects their level of product purchase
- 8

Correlation

```

1 # Let us analyse the numerical columns for thier correlation
2
3 numeric_cols = ['Age','Education','Usage','Income','Fitness','Miles']
4 corr = atm[numeric_cols].corr()
5 corr

```

	Age	Education	Usage	Income	Fitness	Miles
Age	1.000000	0.280496	0.015064	0.513414	0.061105	0.036618
Education	0.280496	1.000000	0.395155	0.625827	0.410581	0.307284
Usage	0.015064	0.395155	1.000000	0.519537	0.668606	0.759130
Income	0.513414	0.625827	0.519537	1.000000	0.535005	0.543473
Fitness	0.061105	0.410581	0.668606	0.535005	1.000000	0.785702
Miles	0.036618	0.307284	0.759130	0.543473	0.785702	1.000000

1 # Fitness and Miles (0.79); usage and Miles (0.76); Fitness and Usage (0.67); Income and education (0.63) these all have a high posit

```

1 plt.figure(figsize=(10, 7))
2
3 sns.heatmap(corr, annot=True, cmap='coolwarm',fmt='.2f',linewidths=0.5)
4
5 plt.title('Correlation Heatmap of Numerical Variables')
6 plt.show()

```

Show hidden output

```

1 # Using Pairplot we can deduce more relations:
2
3 sns.pairplot(atm, vars=numeric_cols, hue='Product', diag_kind='hist')
4
5 plt.show()

```



```

1 # from fitness and Miles pairplot, we can deduce that Customers havinf high fitness rating intend to run/walk longer miles,
2 # hence they prefer advanced featuristic KP781 Product.
3 # Usage and Miles suggest that, customers regularly run/walk each week tend to use advanced treadmill, and this also associated with

```

Probability for Products Purchase

```

1 # Let us see Probability of a Male Customer, who dominates advanced Product category Usage, buying a KP781
2
3 pd.crosstab([atm['Product']], [atm['Gender']], normalize = True)

```

Gender	Female	Male
--------	--------	------

Product

KP281	0.222222	0.222222
KP481	0.161111	0.172222
KP781	0.038889	0.183333

- 1 # The above table shows probabilitiy of gender buying a particular product
- 2 # Now, given That gender is Male let us find out probability of male buying KP781

```
1 pd.crosstab([atm['Product']], [atm['Gender']], normalize='columns', margins = True)
```

Gender	Female	Male	All
--------	--------	------	-----

Product

KP281	0.526316	0.384615	0.444444
KP481	0.381579	0.298077	0.333333
KP781	0.092105	0.317308	0.222222

- 1 # from the above table, it is evident that:
- 2 # for Product KP781 probability of a male buying it is 0.318 or almost 32 %.

Two Way contingency Tables

Product and Gender

```
1
2
3 pd.crosstab(atm['Product'], atm['Gender'], normalize = True)
```

Gender	Female	Male
--------	--------	------

Product

KP281	0.222222	0.222222
KP481	0.161111	0.172222
KP781	0.038889	0.183333

```
1 # Product given Gender
2 pd.crosstab(atm['Product'], atm['Gender'], normalize = 'columns')
```

Gender	Female	Male
--------	--------	------

Product

KP281	0.526316	0.384615
KP481	0.381579	0.298077
KP781	0.092105	0.317308

```
1 # gender given Product
2 pd.crosstab(atm['Product'], atm['Gender'], normalize = 'index')
```

Gender	Female	Male
--------	--------	------

Product

KP281	0.500000	0.500000
KP481	0.483333	0.516667
KP781	0.175000	0.825000

Product and Marital Status

```
1 pd.crosstab(atm['Product'], atm['MaritalStatus'], normalize = True)
```

```
MaritalStatus  Partnered  Single
```

```
Product
```

KP281	0.266667	0.177778
KP481	0.200000	0.133333
KP781	0.127778	0.094444

```
1 # Given Product
2 pd.crosstab(atm['Product'], atm['MaritalStatus'], normalize = 'index')
```

```
MaritalStatus  Partnered  Single
```

```
Product
```

KP281	0.600	0.400
KP481	0.600	0.400
KP781	0.575	0.425

```
1 # Given Product
2 pd.crosstab(atm['Product'], atm['MaritalStatus'], normalize = 'columns')
```

```
MaritalStatus  Partnered  Single
```

```
Product
```

KP281	0.448598	0.438356
KP481	0.336449	0.328767
KP781	0.214953	0.232877

Product and age Group

```
1 pd.crosstab(atm['Product'], atm['Age_Group'], normalize = True)
```

```
Age_Group  Teenage    20s    30s    40s    50s
```

```
Product
```

KP281	0.022222	0.272222	0.111111	0.033333	0.005556
KP481	0.005556	0.177778	0.122222	0.027778	0.000000
KP781	0.000000	0.150000	0.044444	0.027778	0.000000

```
1 # Given Age group
2 pd.crosstab(atm['Product'], atm['Age_Group'], normalize = 'columns')
```

```
Age_Group  Teenage    20s    30s    40s    50s
```

```
Product
```

KP281	0.8	0.453704	0.40	0.3750	1.0
KP481	0.2	0.296296	0.44	0.3125	0.0
KP781	0.0	0.250000	0.16	0.3125	0.0

```
1 # Given Product
2 pd.crosstab(atm['Product'], atm['Age_Group'], normalize = 'index')
```

```
Age_Group  Teenage    20s    30s    40s    50s
```

```
Product
```

KP281	0.050000	0.612500	0.250000	0.075000	0.0125
KP481	0.016667	0.533333	0.366667	0.083333	0.0000
KP781	0.000000	0.675000	0.200000	0.125000	0.0000

Product and Fitness

```
1 pd.crosstab(atm['Product'], atm['Fitness'], normalize = True)
```

Fitness 1 2 3 4 5

Product

KP281	0.005556	0.077778	0.300000	0.050000	0.011111
KP481	0.005556	0.066667	0.216667	0.044444	0.000000
KP781	0.000000	0.000000	0.022222	0.038889	0.161111

```
1 # Given Product
2 pd.crosstab(atm['Product'], atm['Fitness'], normalize = 'index')
```

Fitness 1 2 3 4 5

Product

KP281	0.012500	0.175	0.675	0.112500	0.025
KP481	0.016667	0.200	0.650	0.133333	0.000
KP781	0.000000	0.000	0.100	0.175000	0.725

```
1 # Given Fitness
2 pd.crosstab(atm['Product'], atm['Fitness'], normalize = 'columns')
```

Fitness 1 2 3 4 5 

Product

KP281	0.5	0.538462	0.556701	0.375000	0.064516
KP481	0.5	0.461538	0.402062	0.333333	0.000000
KP781	0.0	0.000000	0.041237	0.291667	0.935484

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- **Customer Profiling**

- 📌 **KP281:**

- KP281 is the most purchased treadmill and attracts a wide range of customers. Most buyers are young adults, with average fitness levels and moderate usage. It is suitable for customers looking for an affordable and basic treadmill.

- 📌 **KP481:**

- KP481 is preferred by customers who want something between the basic and premium models. Its customers are mostly young adults with average fitness levels and regular usage.

- 📌 **KP781:**

- KP781 is mainly purchased by customers who are more serious about fitness. They generally have higher income, higher fitness levels, higher weekly usage and expect to cover more miles. Male customers form a large part of this group.

- **Key Insights**

1. KP281 is the most popular product, with 80 purchases, followed by KP481 with 60 and KP781 with 40.
2. Most AeroFit customers are in their 20s, so young adults form the main customer group.
3. Age does not show a major difference between the three products, so age alone may not be enough to recommend a treadmill.
4. KP781 attracts more fitness-focused customers. Most of its customers have high fitness ratings.
5. KP781 customers plan to use their treadmills more frequently than customers buying KP281 and KP481.
6. KP781 customers expect to cover more miles, showing that the premium treadmill is more popular among customers with higher exercise goals.
7. Income is higher among KP781 customers, suggesting that customers willing to spend more are more likely to choose the premium model.
8. Male customers form a much larger share of KP781 buyers, with 33 male buyers compared with 7 female buyers.
9. Marital status does not show a major difference across the products, as partnered customers are the majority for all three products.
10. The correlation analysis shows that Fitness, Usage and Miles move together strongly, which supports the idea that customers who are more fitness-focused tend to exercise more and expect to cover more distance.

- **Recommendations**

1. Promote KP281 as the beginner and affordable option for customers who want a simple treadmill for regular exercise.
2. Position KP481 as the middle option for customers who want more than a basic treadmill but do not need the premium model.
3. Promote KP781 towards serious fitness customers who exercise frequently and expect to cover more miles.
4. Ask customers about their exercise habits before recommending a product. How often they plan to use the treadmill and how much they plan to exercise can help in choosing the right model.
5. Focus premium promotions on customers who are comfortable spending more, as KP781 buyers generally have higher incomes.
6. Do not depend only on age when recommending a treadmill, because customers across the three products are mostly from a similar age group.
7. Use fitness level and expected usage to guide recommendations. Customers with lower or moderate fitness and usage can be shown KP281/KP481, while frequent and highly fit customers can be shown KP781.

- **Final Conclusion**

The analysis shows that KP281 is the most popular treadmill and attracts a broad range of customers. KP481 serves customers looking for a middle-level option. KP781 has a more specific customer group, mainly higher-income and fitness-focused customers who plan to use the treadmill more often and cover more miles. Therefore, AeroFit can improve its recommendations by understanding the customer's fitness level, exercise habits, expected usage and budget rather than relying only on age or gender.